



Article

Where Do Grocers Go Digital: Spatial Modeling of E-Commerce Configuration Decisions in U.S. Food Retailing

M. Harshvardhan ^{1,*} , Connor Reed ² and Mark P. Sallos ¹

¹ Department of Accounting and Information Systems, School of Business Administration, American University of Sharjah, University City, Sharjah 26666, United Arab Emirates; msallos@aus.edu

² Meijer Companies Ltd., Grand Rapids, MI 49544, USA; connor.reed@meijer.com

* Correspondence: harshvardhan@aus.edu; Tel.: +971-6-515-4644

Abstract

Supermarket chains must decide whether individual stores will offer pickup, delivery, both services, or neither. While consumer preferences for e-commerce features are well studied, the firm-level and store-level decisions that drive channel offerings remain underexplored. We analyze an augmented dataset of market, demographic, spatial and e-commerce information, assembled from three proprietary sources, covering 14,275 grocery stores in the United States. We use a spatial multinomial generalized additive model (GAM) to predict the offered configuration at the store level, benchmarked against elastic net multinomial regression and random forest models, and find that spatial GAM achieved a macro ROC AUC near that of the random forest (0.873 vs. 0.896), while having the strongest balanced accuracy across configurations (0.568). The banner was the most important predictor across all model families, while other important factors include the store type and demographic variables like historical population growth, education levels, population density, household size, income levels and geospatial locations. Our analysis indicates that the local market context retains a significant secondary role that cannot be fully separated from organizational factors in determining the store's policies. Finally, 15.2% of the stores in our dataset have different predicted configurations in models with and without the banner. This identifies stores whose predicted configurations change when banner information is omitted and may therefore warrant closer managerial review.

Keywords: retail location strategy; spatial multinomial GAM; omnichannel retailing; trade area analysis; e-commerce; digital divide



Academic Editors: Nikolaos
T. Giannakopoulos and Damianos
P. Sakas

Received: 4 September 2026

Revised: 5 October 2026

Accepted: 7 October 2026

Published: 10 October 2026

Copyright: © 2026 by the authors.

Licensee MDPI, Basel, Switzerland.

This article is an open access article
distributed under the terms and

conditions of the [Creative Commons
Attribution \(CC BY\) license](#).

1. Introduction

The size of the United States groceries market was projected to grow more than eight-fold over the last ten years, from \$10.6 billion in 2016 to an expected \$89.5 billion by the end of 2026, equivalent to 14% of the total grocery spending [1]. This growth was accelerated by the COVID-19 pandemic, which increased the scope of online services and e-commerce adoption patterns within the sector [2]. The market share of the online segment has also shown signs of consolidation, with Walmart accounting for roughly 36% of online grocery sales nationally as of 2023 [1,3]. The growing indication of market-share momentum effects increases the competitive pressure on supermarkets to re-evaluate offerings [4]. Under these conditions, effective responses must address a critical resource allocation question: given a heterogeneous market with local diversity in conditions, and costs of implementing

changes, which e-commerce model would be most effective in each location? The available choices include pickup, delivery, both, or neither.

Despite the broader trend towards digitalization and e-commerce growth, individual answers are significantly more granular and potentially contentious, as the decision must account for a range of operational and economic implications. Omnichannel grocery models can add significant logistical strain, driven by channel conflicts, higher distribution expenses, lower profit margins, and first-class tracking requirements for order-completion timelines [5]. These tensions are reflected across national and regional U.S. grocery networks that adopted differentiated e-commerce strategies, with different configurations across their stores based on local market needs. Prior research has primarily focused on the consumer behavior aspects of the problem, addressing the factors that determine who purchases through digital channels. The retailer factors and considerations that underpin the supply /offering side of the problem and drive decisions regarding distribution and format remain understudied [6,7].

As noted, these decisions have a competitive dimension [4]. For retail chains, the challenge to keep their offerings updated with market trends intensifies as digitalization becomes the norm. As Novais Filho et al. [8] (p. 466) note, “many organizations struggle to adapt their decision-making processes to the new modes of value creation that digital technologies enable.” In omnichannel systems, Gao and Su [9] documented persistent separation between online and offline operations within many firms. Hu et al. [10] describe a tendency of supermarkets to underutilize analytical capabilities, such as advanced modeling and spatial analysis, at the local level. Thus, given sectoral diversity, many retailers may lack accessible, data-informed methodologies for navigating complex configuration decisions, which positions our work as well suited to addressing an underexplored applied research need.

We address this modeling gap with spatial multinomial models of store-level e-commerce configurations. Our primary contribution is based on a spatial multinomial generalized additive model (GAM) that allows broad, nonlinear spatial variation while retaining demographic, market, and store-format covariates. Additionally, for benchmarking, we use an elastic-net based multinomial regression, which provides a regularized linear benchmark, and a random forest model, which provides a flexible predictive benchmark. Since every specification uses the same observations, common non-banner predictors, regional folds, SMOTE rule, evaluation metrics, and permutation-based importance measure, this design provides us with a single, consistent basis for comparing accuracies of different models as well as the importance of covariates.

Our modeling strategy relies on our second empirical contribution. We integrate three proprietary commercial data sources at the store level: ChainXY (<https://www.chainxy.com/>, accessed on 3 September 2026) records of locations, banners, and e-commerce configurations; SiteWise (<https://www.sitewise.com/>, accessed on 3 September 2026) drive-time trade areas; and Synergos (<https://www.synergos-tech.com>, accessed on 3 September 2026) demographic and market data which connect firm deployment decisions to specific local market characteristics for each store. Building on the data infrastructure introduced by Reed [7], the present study extends the analysis from a binary offering decision to four distinct channel configurations and evaluates their geographic generalizability through out-of-region validation. Distinguishing pickup, delivery, and their joint provision allows us to examine whether market and spatial associations differ across configurations that a binary adoption outcome combines. The spatial GAM represents broad nonlinear geographic variation, while the paired banner specifications and held-out regional evaluation examine how configuration predictions depend on organizational information and generalize across locations.

The rest of this paper proceeds as follows. Section 2 reviews the existing literature on consumer adoption of e-commerce channels, spatial inequality, and firm-level omnichannel capability. Section 3 describes the data and model input features, spatial GAM, benchmark models, and leave-one-region-out validation checks. Section 4 reports comparative and class-specific performance, organizational consistency and regional adaptation, model-agnostic variable importance, and decision-support diagnostics. Then, Section 5 discusses organizational consistency alongside regional adaptation, considering the direction of market and spatial relationships, and states the limitations of our study. Finally, Section 6 concludes our work.

2. Review of Literature and Theoretical Foundations

2.1. Demographic Drivers of E-Commerce Adoption

Substantial empirical research characterizes which consumers adopt online shopping (Table 1). The most common model specification in this literature is regression modeling, particularly discrete choice specifications in which individual or household adoption decisions are regressed on demographic covariates. This approach provides direct interpretability: coefficient directions and sizes indicate which characteristics drive adoption and their strengths. Table 1 below compiles results from several investigations across multiple nations, product sectors, and years, documenting the e-commerce measurement and statistically significant demographic associations reported in each work.

Table 1. Summary of demographic drivers of e-commerce consumption. ✓ means a statistically significant association; ✗ means no statistically significant association; ~ means variable was not part of the study.

Author(s)	Location	E-Commerce Measure	Age	Income	Education	Other Significant Variables
Krizek et al. [11]	US	Adoption; frequency	✓	~	~	Employment; transit access
Punj [12]	US	Adoption	✓	✓	✓	-
Farag et al. [13]	The Netherlands	Adoption; frequency	✓	✓	✓	Employment; household composition
Weltevreden and van Rietbergen [14]	The Netherlands	Adoption	✓	✓	✓	Household size; employment
de Blasio [15]	Italy	Purchase likelihood	✓	✓	✓	PC access; urban/rural
Clarke et al. [16]	UK	Frequency	✓	✓	✓	Vehicle ownership; household size
Van Droogenbroeck and Van Hove [17]	Belgium	Adoption	✓	✓	✓	Household composition; employment
Beckers et al. [18]	Belgium	Adoption	✓	✓	✓	Employment; household composition
Loo and Wang [19]	China	Adoption	✓	✓	✓	Employment; car ownership
Zhen et al. [20]	China	Channel choice	✓	✓	✓	Retail proximity; employment

Table 1. Cont.

Author(s)	Location	E-Commerce Measure	Age	Income	Education	Other Significant Variables
Shao et al. [21]	China	Activity index	✓	✓	✓	Urban/rural; retail accessibility
Saphores and Xu [22]	US	Adoption; expenditure	✓	✓	✓	Employment; household composition
Ganning and Green [23]	US	Online retail expenditure	✓	✓	✓	MSA distance; retail access
Colaço and de Abreu e Silva [24]	Lisbon, Portugal	Adoption (retail + food delivery)	✓	✗	~	Employment status (not in workforce)
Titiloye et al. [25]	Florida, US	Online + in-store grocery frequency	~	~	~	Attitudes (tech savviness, pro-alternative modes, pro-online, joy of shopping, cost consciousness); retired; living alone; no driver's license in HH; product return frequency
Algherbal et al. [26]	Bahrain	Frequency, 4 product categories, 3 periods	~	✗	~	Gender; daily internet use; occupation

We find three consistent patterns across the literature on this front. First, age, income, and education serve as robust adoption predictors across geographic and sectoral boundaries. Across the United States [11,12,23], the Netherlands [13,14], the United Kingdom [16,27], Belgium [17,18], Italy [15], and China [19–21], early adopters often map to a particular demographic profile: younger age, higher income, elevated educational credentials. This regularity is consistent across survey designs, observation periods, and variation in the definition of outcome variables. It is also consistent with early-adopter characteristics documented in different technological shifts, including online banking [28] and transportation-network services [29]. This suggests that e-commerce acceptance follows a generalizable trajectory in which newer technologies first enter markets through younger, wealthier, more highly educated segments.

Second, this demographic regularity is also robust within the food retail literature. Clarke et al. [16] documented that frequent online grocery purchasers in the United Kingdom are younger, wealthier male households with fewer residents. Vehicle ownership was associated with a higher probability of online adoption independent of income controls, implying that time pressures rather than mobility constraints drive channel preferences. Van Droogenbroeck and Van Hove [17] found younger, higher-income Belgian households featuring school-age children and multiple wage-earners to be most receptive to online grocery pickup; this combination is a reflection of both time limitations and logistical pressures faced by the demographic. Pernot [30] documents similar findings in the French market. As groceries constitute a low-discretionary, repeat-purchase category, the consistent role played by the youth–wealth–education profile across food retail indicates that these traits represent fundamental adoption drivers across retail formats.

Third, despite these regularities, there are also inconsistencies and variability in the patterns and relationships presented by demographics across domains. Gender il-

illustrates this variability most clearly, as some analyses find men more prone to online shopping [13,16,18], while others find the opposite [22,31]. Race also presents mixed effects, remaining a comparatively understudied demographic variable in this context. For instance, Saphores and Xu [22] reported lower online grocery adoption among African American consumers. In contrast, Ganning and Green [23] document higher online expenditure in counties with larger Black populations, though they note that regional variation may partly obscure this result. Ren and Kwan [31], in a Columbus study, found white consumers more likely than Black consumers to purchase online even after adjusting for other variables. Yet their sample showed no significant income or age effects, diverging from broader patterns. Such variation suggests that demographic associations operate through local geography, product category specificity, and market-level conditions in manners that conventional additive regression models may not fully capture.

Interestingly, recent studies have shown that income may not have a huge impact on online shopping. Among the recent studies reviewed, income is insignificant in two of them [24,26], and only indirectly affects shopping behavior through consumer attitudes in the third [25]. We posit that as e-commerce has matured, there is a real narrowing of the impact of income effects. In some cases, as Titiloye [25] reported, it may operate through attitudes towards shopping and technology. In other cases, factors like having a driving license in the household [25] and travel time to the nearest grocery store [32] become more important. Inspired by these findings, we define each store's trade area by the drive time and include competitive saturation within that area as a covariate. These modeling choices are covered in more detail in Section 2.2, where we discuss spatial dynamics.

Though focused on consumer choices, previous investigations provide an empirical scaffold for understanding retailer behavior and help us understand why chains target particular markets. Consumer adoption predictors, measurable at the trade-area scale, include the median household income, education level, average age, and family structure, all of which can be derived from standard demographic vendors. Chains calibrating e-commerce capacity to anticipated consumer demand would reasonably assess trade-area demographics. Consequently, the features included in our model (Table 2) draw directly from this established empirical basis.

Table 2. Analysis predictors and full-sample summary statistics.

Feature	Minimum	Maximum	Median	Mean	Standard Deviation
Population Density	9.3	55,356.7	1464.4	2418.3	3738.5
% Population Growth (2010–2020)	−43.6%	415.1%	7.1%	9.2%	11.8%
Median Age	21.4	75.6	38.9	39.3	4.5
Average Household Size	1.6	4.2	2.5	2.5	0.3
Unemployment Rate	0.2	24.4	3.4	3.6	1.2
Median Household Income	\$26,323	\$178,269	\$60,176	\$65,172	\$20,255
Average Number of Vehicles	0.3	2.8	1.8	1.8	0.2
% of Population with a Bachelor's Degree or Higher Education	4.8%	87.6%	31.4%	33.6%	13.5%
% Black Population	0.0%	94.7%	7.9%	12.6%	13.5%
% Hispanic Population	0.4%	96.9%	10.1%	15.9%	16.0%

Table 2. Cont.

Feature	Minimum	Maximum	Median	Mean	Standard Deviation
% Male Population	42.4%	66.3%	48.9%	49.0%	1.2%
Competitive Saturation	0.00	0.89	0.13	0.14	0.06
E-commerce % of Consumer Demand	1.2%	28.0%	9.5%	10.5%	6.4%
Longitude	Longitudinal coordinate system of store location				
Latitude	Latitudinal coordinate system of store location				
Type	Categorical variable representing the type of store (wholesale club/supercenter, specialty grocer, supermarket, or other)				
Banner	Categorical variable representing name of grocery chain				

2.2. Spatial Dynamics: Competing Hypotheses

Spatial conditions shape e-commerce adoption beyond demographic composition. Anderson et al. [33] proposed an early spatial framework for identifying where the digital retail industry would develop, presenting two competing hypotheses. The innovation-diffusion hypothesis posits that cities are adoption pioneers, as urban populations possess higher education levels and greater internet engagement beyond employment. In contrast, the efficiency hypothesis suggests that rural and isolated consumers gain greater benefit from online retail's product breadth, which leads to increased adoption. The two hypotheses yield conflicting predictions—innovation diffusion predicts concentration in high-density metropolitan areas; efficiency predicts accelerated adoption in places where physical retail infrastructure is limited.

Reflecting the two mechanisms, past research operationalizes spatial context through both urbanity and accessibility to physical shopping opportunities. Farag et al. [13] classified residential environments from very strongly urbanized to nonurbanized, and measured accessibility by the number of non-daily-goods shops reachable within specified driving-time bands. Ren and Kwan [31] represented the residential context using the population density and assessed retail accessibility through cumulative-opportunity measures of non-daily shopping. Zhen et al. [20] distinguished urban and suburban home and workplace locations and examined store accessibility using the travel time and travel mode.

Empirical work incorporating spatial or geographic variables employs a range of methods including logistic regression, linear regression, multinomial logit and probit models, and some spatial econometric specifications. Their findings are mixed: several studies report no significant spatial effect on online purchasing behavior [11,14,15,18], while others find support for the efficiency hypothesis, in which consumers in areas with limited retail access turn to e-commerce as a substitute [16,23,31]. A smaller subset supports the innovation-diffusion hypothesis, where denser or more urbanized areas exhibit higher online purchasing rates [13,21,34], and interestingly, Zhen et al. [20] find evidence for both mechanisms operating simultaneously. Kirby-Hawkins et al. [6] similarly find support for both mechanisms under distinct circumstances. However, only two of the eleven studies focus specifically on grocery e-commerce [15,16], which means that the spatial dynamics of online grocery shopping remain underexplored relative to general e-commerce.

The lack of resolution between the two hypotheses probably reflects underlying complexity in spatial processes. de Blasio [15] attributed reluctance in rural populations toward e-commerce to factors beyond accessibility and cost calculations, and found that cultural dimensions are influential for customer behaviors. They also found that restricted exposure

to online purchasing behaviors, infrastructural gaps that affect payment systems, and inconsistent delivery service quality all constrain adoption. Behavioral imitation, central to de Blasio's explanation, depends on observation: rural dwellers witness fewer neighbors making online purchases and thus have reduced experiential opportunities. Ganning and Green [23] address the efficiency mechanism from a counter-perspective, exploring how geographic conditions can still inhibit e-commerce adoption. Specifically, they show that distant, lower-income populations encounter shipping expenses, which make consolidated purchasing during retail visits more economical than placing individual orders.

Wieland [35] proposed (during the COVID-19 pandemic) that the two hypotheses' opposition may be diminishing. Two decades of market consolidation and essentially universal broadband penetration across rural and metropolitan zones are expected to have narrowed the adoption gap. Now, five years later, FMI [36] shows that 80% of US food retailers have some form of groceries sales online, as compared to around 50% before 2020. The segment which still has not adopted e-commerce consists of largely small, independent retailers. Thus, medium constraints and availability drivers are no longer the key supply bottleneck. While both hypotheses seek to explain consumer adoption preferences, neither fully addresses where retailers make these services available.

Demand-side (consumer) analyses leave a question our work seeks to answer: what patterns emerge from actual retailer deployment? If chains place e-commerce disproportionately in urban, wealthy trade areas, this corroborates supply-side innovation diffusion, regardless of whether rural consumers would utilize services if available. Actual deployment constitutes a distinct empirical window on the competing accounts proposed by Anderson et al. [33], potentially clarifying the consumer-scholarship deadlock.

2.3. Firm-Level Omnichannel Capability and Spatial Context

Retail channel configurations follow the organizational strategy alongside local market demand factors. Among U.S. retailers, cross-channel integration depends on information technology capabilities as well as organizational and environmental conditions [37]. Omnichannel operations similarly require aligning technology, processes, resources, and capabilities, a coordination challenge compounded for incumbent firms by legacy systems and established organizational structures [38]. These dynamics shift the focus from individual trade areas to the operating model of the firm serving those areas.

The resulting operational model also has a consumer-facing side. Brand familiarity and cross-touchpoint consistency shape how customers experience omnichannel retailing [39]. Their perceptions of channel integration can strengthen empowerment, which in turn increases trust, satisfaction, and patronage intention [40]. Underlying this consistency are distinct operating choices: pickup and delivery require different distribution paths, inventory arrangements, and fulfillment workflows, and suitable designs that vary with retail context [41]. Thus, we treat the grocery banner as an organizational signature: a compact identifier for the chain's customer promise, channel policy, technology base, fulfillment network, external partnerships, and managerial routines. While the banner does not isolate these mechanisms individually, it captures them collectively as firm identity.

Finally, the digital divide between regions and populations provides a complementary spatial lens. Geographic and demographic disparities in digital access are well documented [42,43], with race, income, education, infrastructure, and local geography shaping access to and use of digital services [44–46]. Grocery channel use is likewise spatially and socio-demographically uneven [47]. In grocery retailing, this divide represents both household-level digital access and supply-side service availability across local markets.

2.4. From Consumer Adoption to Firm Deployment

Existing e-commerce research primarily focuses on consumer adoption patterns. However, an organization's behavior raises a different question: conditional on consumer demand and local market structure, why do retailers make specific channel configurations available? Store-level offerings reflect both trade-area conditions and corporate policies set above the store. A model intended to generalize geographically must therefore be evaluated on regions excluded from fitting.

We model the decisions by individual stores using a spatial multinomial generalized additive model (GAM), with four possible outcomes (pickup only, delivery only, both, neither). Spatial models are well suited for such modeling as they capture geographic variation that latitude or longitude alone may miss. Finally, the additive structure enables explanations beyond a tree model ensemble. To benchmark our approach, we use random forest (RF) and elastic-net multinomial regression models. The random forest provides a flexible tree-ensemble benchmark, while the elastic net provides a regularized linear benchmark without spatial smoothing.

3. Data Overview and Methodology Details

3.1. Data Overview

Our primary data source is a cross-sectional dataset covering 14,278 grocery stores across the United States obtained from ChainXY, a retail location database vendor. Our sample was collected in June 2022 and includes addresses, geographic indicators (region, latitude and longitude), banner/grocery chains, and their respective e-commerce statuses for the 16 major chains operating in the United States. Three stores, part of the "Other" region, were removed as they could not form a viable validation group. Thus, our final dataset uses 14,275 stores. In addition to this cross-sectional information, we supplement our dataset with trade areas that represent each store's influential region, i.e., the geographical area that each store attracts customers from. These were obtained from SiteWise Analytics, and they allow us to draw drive-time isochrones, a representation of areas that are within, say, X minutes of driving time away from a store. Standard industry benchmarks and the authors' professional experience in grocery location analytics informed the isochrone time selection. Walmart, Sam's Club, and BJ's Wholesale Club attract price-conscious customers who are willing to travel the distance to buy groceries in bulk for savings, and thus they received a drive-time distance of 20 min. On the other hand, neighborhood supermarkets such as Aldi, Publix, and Safeway attract the nearby population, and thus we considered a drive-time distance of 10 min for them. Specialty grocery stores, due to their unique positioning in the market, such as Trader Joe's, Whole Foods, and Target, received a 15 min driving time. Finally, we included demographic information from Synergos Technologies for each trade-area polygon thus obtained. Population density and competitive saturation were computed from these demographic data within each drive-time polygon. The drive-time thresholds were selected from retail-format considerations without using stores' e-commerce outcomes; this feature engineering represents the store-specific market context but can induce dependence among the banner, store type, and catchment-derived predictors.

In terms of each store's offering, the proportions varied unevenly across the United States (see Figure 1). Of the 14,275 stores, 11,225 stores offered both pickup and delivery, 1724 only offered pickup options, 1016 offered only delivery, and the remaining 310 offered neither pickup nor delivery options. Our dataset had high class imbalance, which we accounted for with synthetic minority oversampling using the SMOTENC method within each training fold. That is, with the SMOTENC method [48], we upsampled each minority class to 25% of the largest class using information from the five nearest neighbors. For each banner specification, all three model families (GAM, RF and elastic net) used the same

resampling procedure for consistency. The numeric variables were also preprocessed only using the training data summary statistics to prevent data leakage.

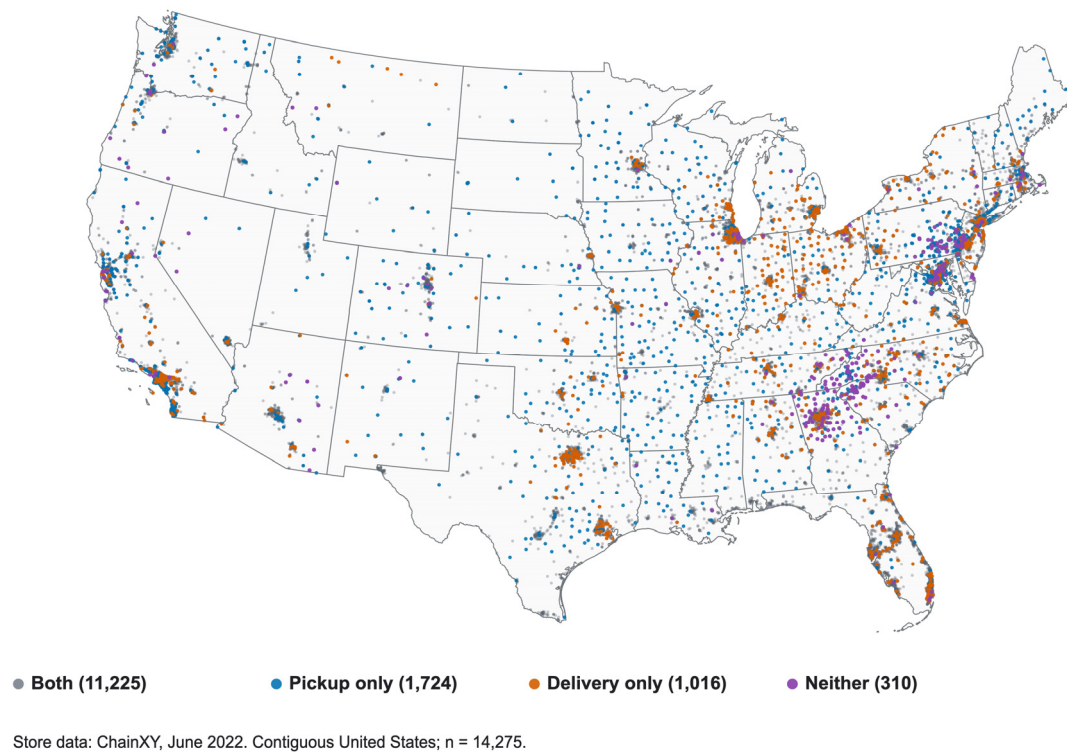


Figure 1. Our study area was the mainland United States, and the map shows the store locations' distribution.

Since banners often dictate their store policies, there is a strong pattern in different banners' offerings of configurations (see Figure 2). Out of 16 original banners in the dataset (including 4 that were compressed to a "regional banner"), 14 had more than 80% of their stores in their most common configuration. ALDI happens to have the strongest variation, with 56.8% of stores having both and 41.3% having delivery only. Similarly, Ingles Markets offers neither pickup nor delivery in 65.1% of stores and pickup only in 34.9% of stores. Thus, while organizations may have a common policy, there are meaningful internal variations within chains.

Preliminary descriptive analysis shows that the supply-side access gap from the retailers cannot be immediately characterized as a demographic divide. While 2.17% of all stores offered neither pickup nor delivery, the share was not the same across population strata: 3.53% of stores in the lowest quartile of population density did not offer pickup or delivery, in contrast to 1.34% in the highest quartile of population density. Similarly, 3.70% of stores in the lowest-income quartile did not offer either service, compared to 1.32% to 2.21% in the remaining quartiles. The shares of stores offering neither pickup nor delivery were similar in the lowest and highest quartiles of Black population share (2.61% and 2.66%, respectively). Across quartiles of Hispanic population share, the no-service share declined from 3.28% in the lowest quartile to 0.90% in the highest. Features used in our model, along with their summary statistics over the entire population, are reported in Table 2 above.



Figure 2. Configuration shares within each original grocery banner.

Since we are using drive-time-based trade-area polygons, we created our own estimates of demographic features. The population density of each polygon represents the number of residents per square mile within each polygon; population growth represents the observed percentage change from 2010 to 2020. Competitive saturation is measured as the number of grocery establishments per 100,000 residents within the drive-time polygons. The share of the local e-commerce demand is calculated independently as the percentage of total consumer spending with e-commerce in the same polygon. For the geographic coordinates (latitudes and longitudes), we project them to the U.S. Albers coordinate system (EPSG:5070) for formulating the spatial term. The store type and banner are used as plain categorical variables. Finally, the BEA economic region [49] is used for constructing validation folds but not directly used as a model feature.

Finally, our model contained 17 features with demographic, market, geographic spatial, and store information. We observed that four banners operated only in a single BEA region and thus were pooled together as a “regional banner”, which prevents a held-out validation fold containing data about stores that are completely absent from the training fold.

3.2. Primary Model and Benchmarks

A spatial multinomial generalized additive model (GAM) is the primary model in our study, which uses a low-rank, two-dimensional smooth projection of geographical location (latitude and longitude) alongside linear transformations for the remaining variables [50]. The spatial basis dimension was selected separately within each outer training sample using the nested regional tuning procedure described in Section 3.3.

To benchmark our primary model, we used two models: elastic-net multinomial regression and random forest. The elastic-net model combines L1 and L2 regularization and thus handles correlation between features effortlessly [51]. The elastic-net mixing parameter and penalty were selected jointly using the nested regional tuning procedure described in Section 3.3. The elastic-net model enables us to have reasonably high predictive power, while restraining ourselves to a linear model. Our other benchmark was a random

forest model [52], which provided a flexible nonlinear comparison; its number of candidate predictors per split, minimum node size, and sampling fraction were selected through the same nested procedure, while the number of trees was fixed at 500.

3.3. Spatial Validation, Hyperparameter Tuning, Metrics, and Importance

To validate our models, we followed a leave-one-region-out design across eight possible regions in our data, i.e., each model was trained on seven regions and was tested on the eighth region. While this design is more costly than a simple random split [53,54], we use it to avoid optimistic estimates of out-of-region performance, as stores in the same region are not independent. For each outer fold, hyperparameters were selected exclusively from the seven-region training sample using three single-region inner validation folds. Standardization and SMOTENC were re-estimated within each inner training split, and the configuration maximizing mean macro one-vs.-rest ROC AUC across the inner validation regions was selected. The candidate grids and the distribution of selected settings across outer folds are reported in Tables S1 and S2. To calculate the metrics from this training/validation design, we averaged them across each held-out region, thus giving them equal weight. For our model evaluation, our primary metric is macro one-vs.-rest ROC AUC [55], in addition to simple accuracy, balanced accuracy and macro F1 as secondary metrics. Table 3 reports the mean and sample standard deviation across the eight held-out regions; dispersion describes regional variability and does not establish statistical superiority from small differences in mean performance.

Table 3. Performance across eight leave-one-region-out test folds, reported as mean $\pm$ sample standard deviation. Macro ROC AUC is the primary metric; regions receive equal weight.

Model	Banner	ROC AUC	Accuracy	Balanced Accuracy	Macro F1
Random forest	With	0.896 $\pm$ 0.054	0.859 $\pm$ 0.038	0.544 $\pm$ 0.072	0.551 $\pm$ 0.076
Spatial multinomial GAM	With	0.873 $\pm$ 0.060	0.819 $\pm$ 0.047	0.568 $\pm$ 0.060	0.533 $\pm$ 0.060
Elastic-net multinomial	With	0.872 $\pm$ 0.038	0.786 $\pm$ 0.079	0.531 $\pm$ 0.112	0.486 $\pm$ 0.096
Random forest	Without	0.834 $\pm$ 0.077	0.809 $\pm$ 0.052	0.455 $\pm$ 0.085	0.456 $\pm$ 0.076
Elastic-net multinomial	Without	0.797 $\pm$ 0.070	0.718 $\pm$ 0.055	0.408 $\pm$ 0.080	0.375 $\pm$ 0.059
Spatial multinomial GAM	Without	0.788 $\pm$ 0.063	0.701 $\pm$ 0.061	0.417 $\pm$ 0.050	0.365 $\pm$ 0.047

Since a banner's corporate policy can sometimes determine e-commerce availability, we estimate each model twice, once using all features and once after removing the banner from the list. (We also test this organizational and regional consistency using its observed mix and configuration share.) To measure each variable's importance, we used permutation importance [56] as the decrease in macro ROC AUC, permuting one feature at a time in each held-out test region—except for projected coordinates, which were permuted jointly to represent common spatial information. Consequently, the fold-level importances were averaged with store counts as weights. We ought to note that while this metric is comparable across model families and measures the predictive power of features, it only represents association and not causality between variables. The banner's importance is not directly comparable to that of individual covariates because it condenses many representations of corporate information into a single feature. Additionally, these importance scores should not be interpreted as independent contributions to the model.

For completeness, we also report one-versus-rest AUC, recall, precision and F1 scores for each model and outcome. For directional assertions, we consider only the store type and the five features supported by the prior literature: population density, observed population growth from 2010 to 2020, percentage of population with a bachelor's degree, competitive saturation, and local e-commerce demand (see Section 2 above). For them, the fitted GAM varies one predictor from the 5th to the 95th percentile while keeping all other characteristics of the store same in each held-out region. The corresponding probabilities are adjusted for the class-prevalence shift introduced by resampling and are averaged within regions.

Because using training-only SMOTENC alters class prevalence due to rebalancing, the predicted probabilities for cross-fits have to be corrected within each fold using the ratio of each class's original training-fold occurrence to its after-SMOTENC occurrence and then renormalized. This probability reliability in our study was measured using three complementary metrics: (1) the multiclass Brier score, which measures the mean squared difference between predicted probability and actual proportions, (2) log loss, which penalizes low probabilities assigned to actual, observed outcomes, and (3) the macro expected calibration error, which measures the absolute difference between binned predicted probabilities and observed proportions. Lower values in these metrics indicate higher reliability.

Model fitting and hyperparameter tuning were conducted in R 4.5.2; Python 3.13.5 was used for diagnostic summaries and figures. Detailed software versions, implementation packages, and reproducibility details are provided in the Supplementary Materials.

4. Results

4.1. Out-of-Region Model Performance

Our spatial GAM is compared with the benchmarks, with and without a banner, in Table 3 and Figure 3. With the banner included, random forest achieves the highest discrimination (macro ROC AUC of 0.896), followed by spatial GAM at 0.873 and elastic net at 0.872. GAM produces a higher balanced accuracy than random forest (0.568 vs. 0.544), while random forest produces a higher macro F1 (0.551 vs. 0.533). Regionally, the GAM accuracy varies from 0.740 in Plains to 0.874 in Southwest. The model families therefore exhibit different predictive strengths, while the reported standard deviations show substantial variation across held-out regions.

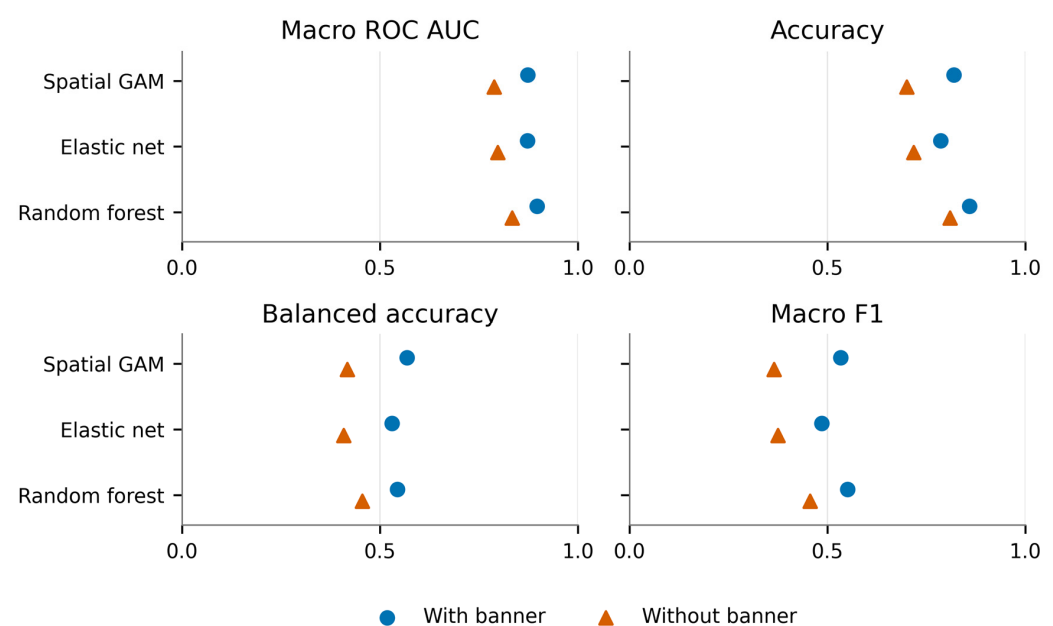


Figure 3. Out-of-region model performance with and without grocery banner.

These aggregate measures conceal differences in class-specific performance. The banner-inclusive spatial GAM has one-versus-rest AUC values of 0.926, 0.910, 0.850, and 0.806 for delivery only, pickup only, both and none, respectively. Using uncorrected held-out predictions, recall is 0.888, 0.633, 0.540, and 0.212 for both, delivery only, pickup only and none, respectively, averaged equally across the eight regions. Low recall for the rarest class limits the model's usefulness for identifying stores that offer neither service, even when overall discrimination is strong. This distinction illustrates why accuracy alone is insufficient to evaluate an imbalanced multiclass model.

4.2. Organizational Consistency and Regional Adaptation

Without the banner, the mean macro ROC AUC is lower for every model. For spatial GAM, it falls by 0.085, from 0.873 to 0.788; corresponding reductions for elastic net and random forest are 0.075 and 0.062. These paired specifications use the same observations and regional validation protocol, supporting the predictive relevance of banner as a proxy for organizational information. Regional changes in GAM macro ROC AUC after adding the banner are Mideast, +0.226; New England, +0.139; Rocky Mountain, +0.123; Far West, +0.081; Great Lakes, +0.069; Southwest, +0.055; Plains, +0.032; and Southeast, −0.047 (Figure 4). This variation may partly reflect the regional banner composition rather than the number of banners represented in each region. The benchmark models also show an overall performance gain from including the banner.

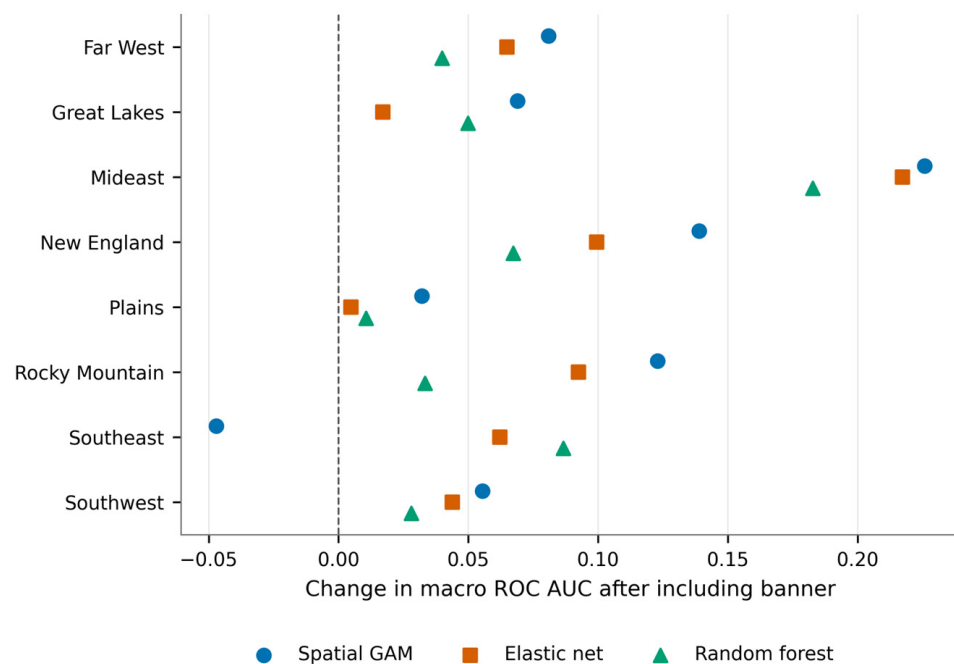


Figure 4. Within-region change in ROC AUC after including banner. Values compare paired specifications using the same held-out region; positive values favor the banner-inclusive model.

4.3. Variable Importance

For spatial GAM, the banner has a permutation importance of 0.260, followed by the store type and demographic variables (Figure 5). Banner importance levels in elastic net and random forest are 0.188 and 0.198, respectively. In all three models without the banner, the store type is the dominant predictor, with mean AUC reductions ranging from 0.231 to 0.247.

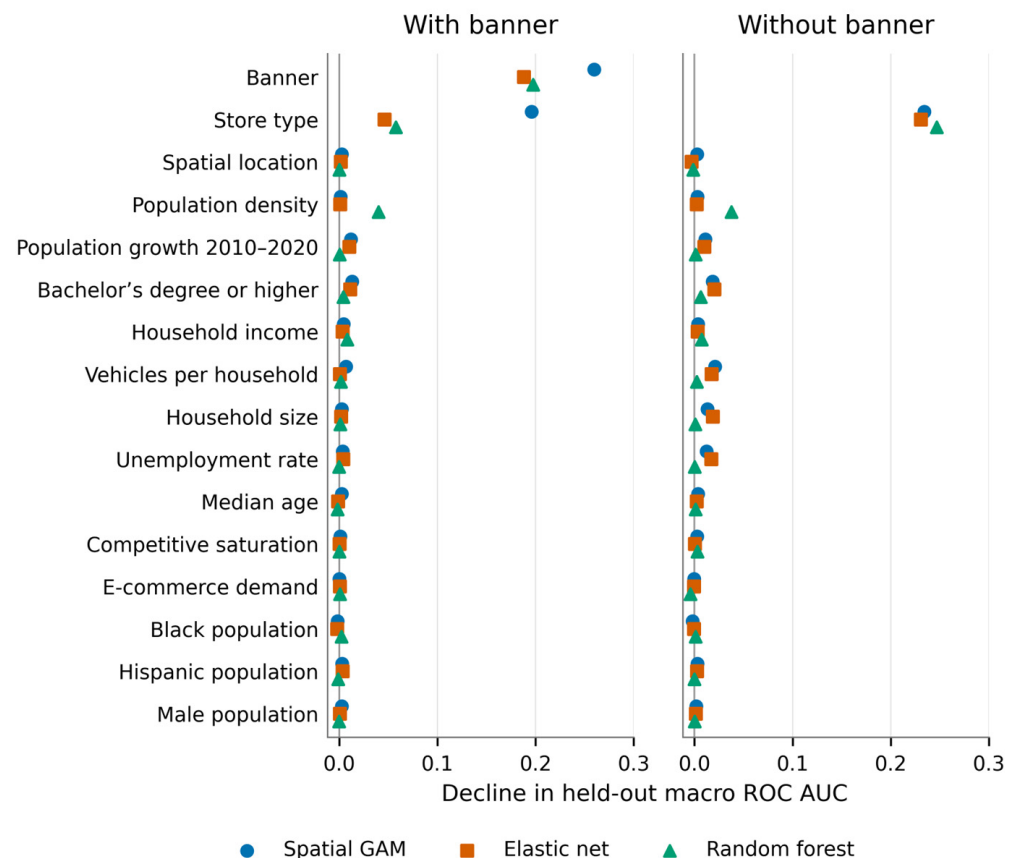


Figure 5. Model-agnostic permutation importance measured as the decline in held-out macro ROC AUC. Spatial coordinates are permuted jointly.

The prominence of the banner and store type indicates the predictive relevance of organizational and format information, but their importance scores do not separate their independent contributions.

The remaining predictors contribute smaller decreases in held-out macro ROC AUC. Spatial multinomial GAM assigns the largest remaining importance scores to a bachelor's degree or higher, population growth (2010–2020), and the average number of vehicles. The elastic-net multinomial assigns the largest remaining importance scores to a bachelor's degree or higher, population growth (2010–2020), and the unemployment rate. Random forest assigns the largest remaining importance scores to population density, median household income, and a bachelor's degree or higher. These scores describe predictive associations rather than independent effects. Population growth in all models refers only to the observed 2010–2020 measure.

4.4. Directional Response Profiles

Cross-fitted response profiles show the direction of the fitted associations while retaining held-out stores' other characteristics. In the GAM with a banner, increasing the share with a bachelor's degree from its 5th to 95th percentile changes the prior-adjusted probability of both services from 0.739 to 0.877. Over the corresponding range, historical population growth changes this probability from 0.758 to 0.864, and population density changes it from 0.811 to 0.765 (Figure 6).

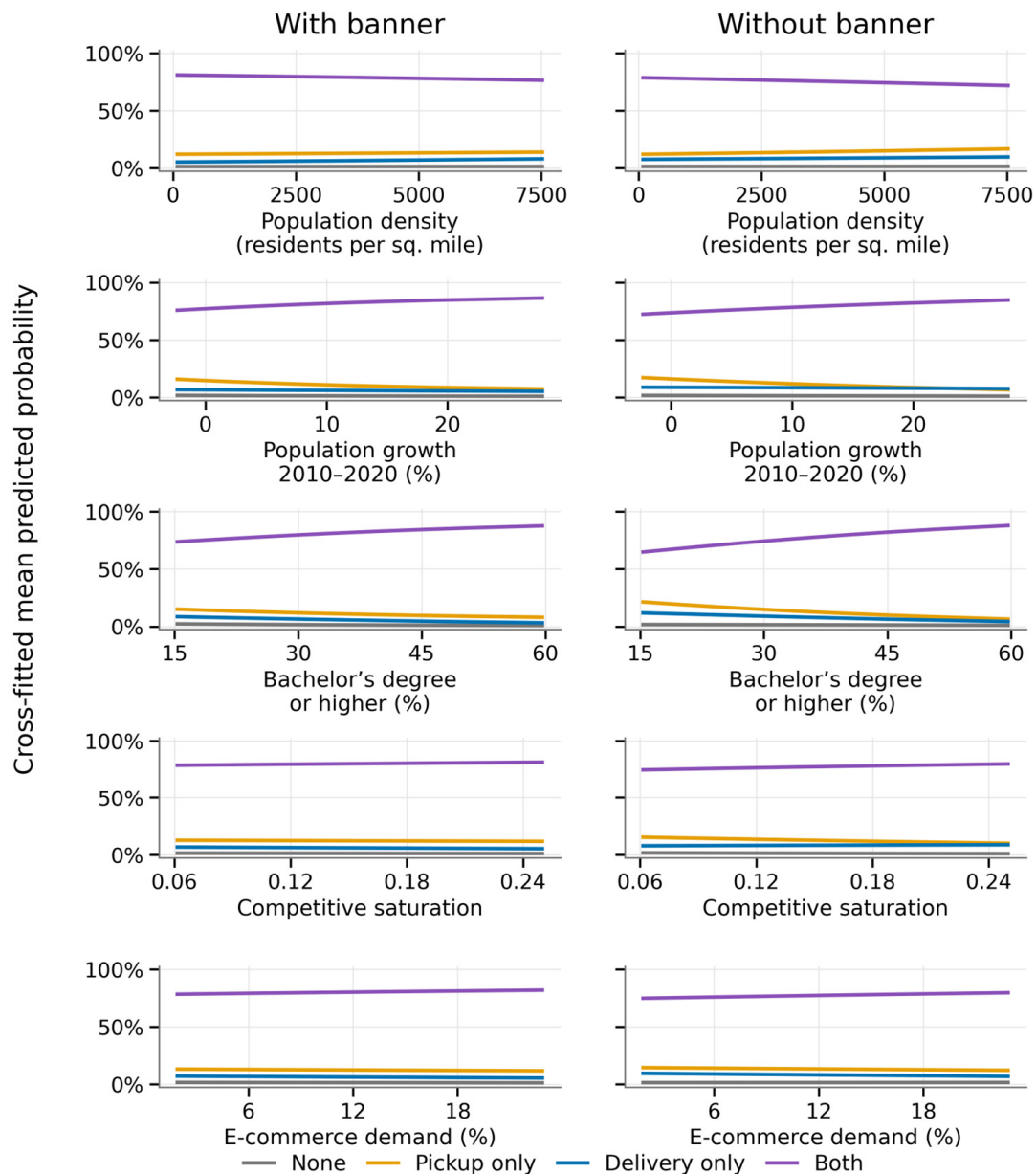


Figure 6. Cross-fitted, prior-adjusted GAM response profiles for theoretically selected market variables. Each curve varies one predictor while retaining held-out stores' other observed characteristics; probabilities are averaged equally across regions.

Without a banner, the probability of both services changes across the 5th–95th percentile range by +23.0 percentage points for education, +12.4 percentage points for historical population growth, +4.8 percentage points for local e-commerce demand, +5.1 percentage points for competitive saturation, and −6.9 percentage points for population density. The store format yields a differentiated profile (Figure 7). The banner-excluded GAM assigns probabilities of both services of 0.904 for clubs and supercenters and 0.848 for the Other category. Specialty stores have probabilities of 0.613 for pickup only and 0.321 for both. Supermarkets have probabilities of 0.664 for both and 0.222 for delivery only. These profiles quantify the predictive associations underlying store type's importance.

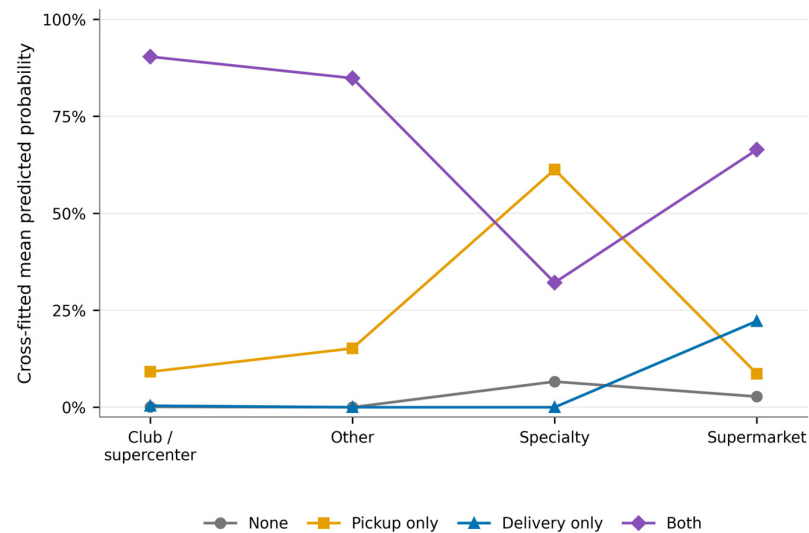


Figure 7. Prior-adjusted store-format response profile from the banner-excluded spatial GAM. Other observed characteristics are retained, and probabilities are averaged equally across held-out regions.

Class-prevalence adjustment corrects the probability shift introduced by training-fold SMOTENC. For GAM with the banner, the multiclass Brier score changes from 0.258 to 0.216, log loss from 0.600 to 0.550, and macro expected calibration error from 0.054 to 0.026. Corresponding metrics for all model families and both banner specifications appear in Table S4. After adjustment, GAMs with and without the banner assign different configurations to 15.2% of stores. The highest disagreement occurs in the Far West (24.3%) and Mideast (23.5%), consistent with regional variation in the role of banner information.

5. Discussion and Limitations

5.1. The Banner as an Organizational Signature

Our strongest result is the high predictive power of the banner. The banner is a useful proxy for the chain's organizational signature, which identifies its customer promise, channel policy, technical architecture, fulfilment network and reach, external relationships and managerial routines. This is a consistent story that is repeated across chains. Considering the gains with the inclusion of the banner in geographically held-out data, we can see how these firm-level choices can travel across local markets.

For companies trying to have a consistent brand image across their channels, with an omnichannel strategy, this explanation particularly fits as cross-channel integration depends on retailer IT capabilities and organizational conditions [37]. Such omnichannel execution also requires modifications in technology and capabilities, especially with legacy systems [38]. For customers, this strategy reinforces brand familiarity and a consistent cross-channel experience [39]. Their perceptions of integration also increase trust and satisfaction with the brand [40]. Even though different operating configurations like pickup and delivery have very different requirements for execution, having a single corporate policy, represented as the banner in the model, can carry these elements all at once.

The importance of the banner also shows that many fulfilment decisions may be made at the corporate level and not at the store level. It may still be dependent on the type of store, which is consistent with some organizations making these decisions above the store level. Having standardized websites and apps, shared inventories, and fulfilment promise across brands might be driving store-level decisions. Thus, while local trade-area characteristics are important, they often operate within a framework that is designed by the parent organization. This is further corroborated by the fact that 14 out of 16 banners have modal configuration shares above 80%, showing that the conclusions are not limited

to a few national chains. Furthermore, the regional variation may be explained by the composition of chains, each company's distribution capacity and operating constraints. Retailers who have a strong, profitable presence in a market may choose to experiment with their configuration of having delivery and pickup that they do not otherwise have nationally. These variations may also arise from internal governance differences, i.e., differences in how managers from different regions approach the solution with different levels of capital, labor and technological support at their disposal. For practical purposes, organizational consistency and regional adaptations may be complementary methods for a chain's success in the market.

5.2. Interpreting the Other Important Factors

After the banner, the store type is the strongest predictor of the configuration. Different types of stores, such as supercenters, wholesale clubs, specialty grocers, and supermarkets, vary by customer footprint, available assortment, labor force staffing, inventory management and, of course, order fulfillment. Following the parallelism that the feature banner captures the nationwide organizational policy, the store type captures local customer and company behavior. This is also directly related to the distribution design problem, as noted by Hübner et al. [41].

The remaining important features describe the market's characteristics, like demographics. For instance, population density determines both the concentration of potential orders and the distribution of last-mile delivery operations. Similarly, education, income, vehicle availability, household size and population growth are representations of e-commerce readiness, financial capacity, travel options, demand size and potential for market expansion, respectively. Spatial features, integral to our GAM, capture the regional conditions, as some areas have geographies supporting characteristics such as labor supply, infrastructure availability, legal environment and supply chain readiness. These features speak to the plausible mechanisms of consumer adoption from the supply side.

The cross-fitted response profiles show the direction of these demand-side associations. For instance, a higher share of residents with a bachelor's degree or higher and greater population growth is associated with a greater probability of dual-service deployment, which is consistent with the innovation-diffusion hypothesis. A higher population density modestly shifts the probability away from a both-services configuration, consistent with the efficiency hypothesis, which posits that online channels have greater value in areas where consumers are geographically dispersed. Taken together, these findings revalidate Anderson et al.'s [33] competing hypotheses theory.

Although trade areas with a lower population density and lower income have a disproportionate share of no-service stores, the GAM response profiles associate an increasing population density with a declining probability of both services. Smaller positive associations of local e-commerce demand further suggest that chains respond to market viability in addition to geographical suitability. Thus, we conclude that both innovation-diffusion and efficiency mechanisms can often coexist under different circumstances [6,20].

When viewed through the lens of the digital divide between two broad groups (rich or poor, race groups, etc.), the descriptive results show that the story may not hold strongly. Although no-service stores were more common in low-density and low-income trade areas, their prevalence did not increase consistently with the Black population share and declined across quartiles of the Hispanic population share. Thus, this divide may result from a sparser market deployment by the chains. These associations should be interpreted jointly with the banner and store type because the catchment-derived predictors share a feature-engineering design informed by the retail format.

5.3. Managerial Implications

The two versions of our GAM (with and without the banner) estimate and answer two related but different questions. Models with the banner predict the fulfilment configuration that is most directly consistent with a chain's established operating scheme for a given trade-area polygon. On the other hand, models without the banner estimate the fulfilment configuration associated with the measured local market characteristics and store type, while omitting the explicit banner identifier. Because the store type remains in the model and is associated with the banner, this specification does not isolate an independent effect of local market conditions. Consequently, differences between the two models identify stores where predicted configurations change when banner information is omitted and provide a diagnostic for managerial review alongside constraints such as capacity, contracts, and capital availability.

With these predictions for the models, a mixed banner profile, a smaller regional gain or a large gap between the two GAM probability vectors for the four configurations all suggest a closer review by a subject matter expert. Managers can review the store format and directional profile to examine why the banner-informed prediction differs from the configuration associated with measured local market conditions. The 15.2% disagreement that we observed between the two GAM specifications is not a list of definite policy changes; rather, we suggest treating the flagged stores as a set of locations for further review by the store managers and/or the chains themselves.

5.4. Limitations and Future Research

Since our dataset is cross-sectional and national in scope, and we do not observe the launch timing, service quality, delivery fees, order volume, fulfillment costs, margins, profitability, capacity, or managerial intent, the model differences should be interpreted as diagnostic signals for review rather than prescriptions for changing store configurations. Even though we used three different high-quality proprietary datasets, all of them describe a single point in time (i.e., are not panel data). Longitudinal data about each store's channel launches, coverage, infrastructure investment, and profitability would allow future research to decompose the impact of the organizational policy (currently absorbed by the banner) into its individual characteristics.

Additionally, as response profiles vary one predictor at a time and are averaged over held-out stores, they describe out-of-sample conditional predictions. Subsequently, our results are not causal estimates, since we lack counterfactual interventions. This limitation is also the result of the fact that we cannot separate our individual inputs to the model as they are not independent of each other. The catchment polygons were constructed using 10, 15, or 20 min drive-time thresholds assigned by retail format, which can induce dependence among the store type, banner, and catchment-derived predictors. Robustness to alternative polygon boundaries or a common drive-time threshold has not been tested because the required alternative polygons and demographic extracts are unavailable in the current analytical dataset.

The low recall for stores offering neither pickup nor delivery limits the model's use as a screening tool for no-service locations. Thus, we posit that predictions should be supplemented with direct information on service availability when assessing potentially underserved retail locations.

6. Concluding Remarks

Our research study reframes the grocery chain's e-commerce availability at the store level as a spatial modeling problem using an integrated dataset of 14,275 grocery stores in the United States. We model four possible outcomes (pickup, delivery, both, or none) using

three models, with a spatial generalized additive model (GAM) as the primary model and elastic net and random forest models as benchmarks using store, trade area, geographic and demographic features. We find that our GAM achieves a mean held-out ROC AUC of 0.873, only slightly behind the highly complex random forest at 0.896, while having a higher balanced accuracy and a more interpretable account of organizational, market and geographic variations.

Three major findings emerge from our study. First, chain identity and store format are the dominant features in predicting a store's observed configuration across geographies. The banner, which we interpret as a summary of the chain's identity and customer promise, results in a higher mean ROC AUC across all three model families, although the gains vary across regions. Second, these gains vary across regions and banners, suggesting that a consistent national policy does not always align with the configuration associated with measured local market conditions. In models without the banner, the store type becomes the dominant factor, followed by the trade area's demographic, market, and geospatial characteristics. Third, these deployment decisions do not reduce to a single urban/rural, income, or racial divide. Rather, different configuration decisions may result from different combinations of anticipated demand, accessibility and organizational capability.

For managers, our recommendation is to consider configuration deployment decisions at two levels: banner-level corporate policy and the configuration associated with measured local market conditions. Although a chain's overall policy may suggest or require a particular configuration, the banner-excluded model provides a comparison based on measured local market conditions. Given that the banner-inclusive and banner-excluded models assign different predicted configurations to 15.2% of stores, we interpret this disagreement as a diagnostic for discussion rather than an automatic recommendation from our model.

Supplementary Materials: The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/jtaer21100357/s1>, Table S1: Nested validation design and candidate hyperparameter grids; Table S2: Frequency of selected hyperparameters across the eight outer regional folds; Table S3: Final tuned out-of-region performance, reported as mean $\pm$ sample standard deviation across eight equally weighted outer regions.; Table S4: Calibration metrics before and after class-prevalence adjustment.

Author Contributions: Conceptualization, C.R.; methodology, C.R. and M.H.; software, M.H.; validation, M.P.S.; formal analysis, M.H.; investigation, M.H.; data curation, C.R.; writing—original draft preparation, C.R. and M.H.; writing—review and editing, M.P.S.; funding acquisition, M.H. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the American University of Sharjah, grant number SB2602, awarded to M.H. The funders had no role in the conduct of this study.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data analyzed in this study were obtained under commercial licenses from ChainXY, SiteWise Analytics, and Synergos Technologies. Because the applicable licenses prohibit redistribution, the data are not publicly available from the authors. Access may be requested directly from the respective data providers, subject to their licensing terms and conditions.

Acknowledgments: We acknowledge the selective use of LLMs (Claude Opus 4.8 and ChatGPT Sol 5.6) as part of the paper's editing process. The tools were used for copyediting, occasional structural compression of ideas, and adversarial reviews of specific sections. Outputs were verified, further edited, and only selectively implemented. The authors take full responsibility for the originality and accuracy of the included work in its entirety.

Conflicts of Interest: Author Connor Reed was employed by the company Meijer Companies Ltd., U.S.A. The remaining authors declare that this research was conducted in the absence of any commercial or financial relationships that could be construed as potential conflicts of interest.

References

- Kong, X.; Çakır, M.; Li, Q. Growth of Omnichannel Grocery Retailing and Food Prices. *Agribusiness* **2025**, *42*, 1816–1830. [CrossRef]
- Donthu, N.; Gustafsson, A. Effects of COVID-19 on business and research. *J. Bus. Res.* **2020**, *117*, 284–289. [CrossRef]
- Moran, C.D. Walmart's E-Grocery Sales Share Hits Record High. *Grocery Dive*, 30 August 2023. Available online: <https://www.grocerydive.com/news/walmarts-e-grocery-sales-share-hits-record-high/692236/> (accessed on 3 October 2026).
- Akturk, M.S.; Ketzenberg, M. Exploring the Competitive Dimension of Omnichannel Retailing. *Manag. Sci.* **2022**, *68*, 2732–2750. [CrossRef]
- Milligan, S.; Davies, I.; Yalabik, B.; Celik, M.; Squire, B. Evolutions of Omni-Channel Fulfillment Performance: An In-Depth Case Study in Grocery Retailing. *J. Oper. Manag.* **2025**, *71*, 651–669. [CrossRef]
- Kirby-Hawkins, E.; Birkin, M.; Clarke, G. An investigation into the geography of corporate e-commerce sales in the UK grocery market. *Environ. Plan. B Urban Anal. City Sci.* **2019**, *46*, 1148–1164. [CrossRef]
- Reed, C. Identifying the determinants of E-commerce offerings among grocery store locations in the United States: A spatial econometric analysis. *Appl. Geogr.* **2024**, *172*, 103418. [CrossRef]
- Novais Filho, M.J.d.; Scur, G.; de Mattos, C.A. Exploring business practices, strategies and value creation through digital transformation in retail companies. *Digit. Transform. Soc.* **2025**, *4*, 465–480. [CrossRef]
- Gao, F.; Su, X. Omnichannel Retail Operations with Buy-Online-and-Pick-up-in-Store. *Manag. Sci.* **2017**, *63*, 2478–2492. [CrossRef]
- Hu, H.; Tan, D.; Thaichon, P.; Wang, B.; Zhu, Z. Grid-based market sales forecasting for retail businesses using automated machine learning and geospatial intelligence. *Expert Syst. Appl.* **2025**, *284*, 127869. [CrossRef]
- Krizek, K.J.; Li, Y.; Handy, S.L. Spatial attributes and patterns of use in household-related information and communications technology activity. *Transp. Res. Rec.* **2005**, *1926*, 252–259. [CrossRef]
- Punj, G. Effect of consumer beliefs on online purchase behavior: The influence of demographic characteristics and consumption values. *J. Interact. Mark.* **2011**, *25*, 134–144. [CrossRef]
- Farag, S.; Weltevreden, J.; van Rietbergen, T.; Dijst, M.; van Oort, F. E-Shopping in the Netherlands: Does Geography Matter? *Environ. Plan. B Plan. Des.* **2006**, *33*, 59–74. [CrossRef]
- Weltevreden, J.W.J.; Rietbergen, T.V. E-shopping versus city centre shopping: The role of perceived city centre attractiveness. *Tijdschr. Econ. Soc. Geogr.* **2007**, *98*, 68–85. [CrossRef]
- De Blasio, G. Urban-rural differences in Internet usage, e-commerce, and e-banking: Evidence from Italy. *Growth Change* **2008**, *39*, 341–367. [CrossRef]
- Clarke, G.; Thompson, C.; Birkin, M. The emerging geography of e-commerce in British retailing. *Reg. Stud. Reg. Sci.* **2015**, *2*, 371–391. [CrossRef]
- Van Droogenbroeck, E.; Van Hove, L. Adoption of Online Grocery Shopping: Personal or Household Characteristics? *J. Internet Commer.* **2017**, *16*, 255–286. [CrossRef]
- Beckers, J.; Cárdenas, I.; Verhetsel, A. Identifying the geography of online shopping adoption in Belgium. *J. Retail. Consum. Serv.* **2018**, *45*, 33–41. [CrossRef]
- Loo, B.P.Y.; Wang, B. Factors associated with home-based e-working and e-shopping in Nanjing, China. *Transportation* **2018**, *45*, 365–384. [CrossRef]
- Zhen, F.; Du, X.; Cao, J.; Mokhtarian, P.L. The association between spatial attributes and e-shopping in the shopping process for search goods and experience goods: Evidence from Nanjing. *J. Transp. Geogr.* **2018**, *66*, 291–299. [CrossRef]
- Shao, R.; Derudder, B.; Witlox, F. The geography of e-shopping in China: On the role of physical and virtual accessibility. *J. Retail. Consum. Serv.* **2022**, *64*, 102753. [CrossRef]
- Saphores, J.D.; Xu, L. E-shopping changes and the state of e-grocery shopping in the US—Evidence from national travel and time use surveys. *Res. Transp. Econ.* **2021**, *87*, 100864. [CrossRef]
- Ganning, J.P.; Green, T. The online marketplace: Zero-order city or new source of social inequality? *Growth Change* **2021**, *52*, 1251–1270. [CrossRef]
- Colaço, R.; de Abreu e Silva, J. Exploring the e-shopping geography of Lisbon: Assessing online shopping adoption for retail purchases and food deliveries using a 7-day shopping survey. *J. Retail. Consum. Serv.* **2022**, *65*, 102859. [CrossRef]
- Titiloye, I.; Sarker, M.A.A.; Jin, X.; Watts, B. Investigating e-grocery shopping behavior and its travel effect. *Int. J. Transp. Sci. Technol.* **2024**, *13*, 91–105. [CrossRef]
- Algherbal, E.A.; Hijazi, H.I.; Al-Ahmadi, H.M.; Abdullah, M. Analysis of changes in in-store and online shopping frequencies due to the COVID-19 pandemic: A case study from Bahrain. *Sustainability* **2024**, *16*, 4996. [CrossRef]

27. Soopramanien, D.G.R.; Robertson, A. Adoption and usage of online shopping: An empirical analysis of the characteristics of “buyers” “browsers” and “non-internet shoppers”. *J. Retail. Consum. Serv.* **2007**, *14*, 73–82. [CrossRef]
28. Lee, M.S.Y.; McGoldrick, P.J.; Keeling, K.A.; Doherty, J. Using ZMET to Explore Barriers to the Adoption of 3G Mobile Banking Services. *Int. J. Retail Distrib. Manag.* **2003**, *31*, 340–348. [CrossRef]
29. Dias, F.F.; Lavieri, P.S.; Garikapati, V.M.; Astroza, S.; Pendyala, R.M.; Bhat, C.R. A behavioral choice model of the use of car-sharing and ride-sourcing services. *Transportation* **2017**, *44*, 1307–1323. [CrossRef]
30. Pernot, D. Internet shopping for everyday consumer goods: An examination of the purchasing and travel practices of click and pickup outlet customers. *Res. Transp. Econ.* **2021**, *87*, 100817. [CrossRef]
31. Ren, F.; Kwan, M.P. The impact of geographic context on e-shopping behavior. *Environ. Plan. B Plan. Des.* **2009**, *36*, 262–278. [CrossRef]
32. Titiloye, I.; Sarker, M.A.A.; Jin, X.; Watts, B. Examining channel choice preferences for grocery shopping during the COVID-19 pandemic. *Int. J. Transp. Sci. Technol.* **2024**, *14*, 57–72. [CrossRef]
33. Anderson, W.P.; Chatterjee, L.; Lakshmanan, T.R. E-commerce, transportation, and economic geography. *Growth Change* **2003**, *34*, 415–432. [CrossRef]
34. Farag, S.; Krizek, K.J.; Dijst, M. E-shopping and its relationship with in-store shopping: Empirical evidence from the Netherlands and the USA. *Transp. Rev.* **2006**, *26*, 43–61. [CrossRef]
35. Wieland, T. Spatial shopping behavior in a multi-channel environment: A discrete choice model approach. *REGION* **2021**, *8*, 1–27. [CrossRef]
36. FMI—The Food Industry Association. FMI’s 77th Annual Food Industry Analysis Finds Significant Investments in Tech, Enhancing In-Store Experience. FMI, 7 July 2026. Available online: <https://www.fmi.org/newsroom/news-archive/view/2026/07/07/fmi-s-77th-annual-food-industry-analysis-finds-significant-investments-in-tech-enhancing-in-store-experience> (accessed on 3 October 2026).
37. Cao, L.; Li, L. Determinants of retailers’ cross-channel integration: An innovation diffusion perspective on omni-channel retailing. *J. Interact. Mark.* **2018**, *44*, 1–16. [CrossRef]
38. Lehrer, C.; Trenz, M. Omnichannel business. *Electron. Mark.* **2022**, *32*, 687–699. [CrossRef]
39. Hickman, E.; Kharouf, H.; Sekhon, H. An omnichannel approach to retailing: Demystifying and identifying the factors influencing an omnichannel experience. *Int. Rev. Retail Distrib. Consum. Res.* **2020**, *30*, 266–288. [CrossRef]
40. Zhang, M.; Ren, C.; Wang, G.A.; He, Z. The impact of channel integration on consumer responses in omni-channel retailing: The mediating effect of consumer empowerment. *Electron. Commer. Res. Appl.* **2018**, *28*, 181–193. [CrossRef]
41. Hübner, A.; Holzapfel, A.; Kuhn, H. Distribution systems in omni-channel retailing. *Bus. Res.* **2016**, *9*, 255–296. [CrossRef]
42. Malecki, E.J.; Moriset, B. *The Digital Economy: Business Organization, Production Processes and Regional Developments*; Routledge: London, UK, 2008. [CrossRef]
43. Warf, B. Contemporary digital divides in the United States. *Tijdschr. Econ. Soc. Geogr.* **2013**, *104*, 1–17. [CrossRef]
44. Forman, C.; Goldfarb, A.; Greenstein, S. Geographic location and the diffusion of Internet technology. *Electron. Commer. Res. Appl.* **2005**, *4*, 1–13. [CrossRef]
45. Ono, H.; Zavodny, M. Race, Internet usage, and E-commerce. *Rev. Black Polit. Econ.* **2002**, *30*, 7–22. [CrossRef]
46. van Deursen, A.J.A.M.; van Dijk, J.A.G.M.; ten Klooster, P.M. Increasing inequalities in what we do online: A longitudinal cross section analysis of Internet activities among the Dutch population (2010 to 2013) over gender, age, education, and income. *Telemat. Inform.* **2015**, *32*, 259–272. [CrossRef]
47. Hood, N.; Urquhart, R.; Newing, A.; Heppenstall, A. Sociodemographic and spatial disaggregation of e-commerce channel use in the grocery market in Great Britain. *J. Retail. Consum. Serv.* **2020**, *55*, 102076. [CrossRef]
48. Chawla, N.V.; Bowyer, K.W.; Hall, L.O.; Kegelmeyer, W.P. SMOTE: Synthetic Minority Over-sampling Technique. *J. Artif. Intell. Res.* **2002**, *16*, 321–357. [CrossRef]
49. Bureau of Economic Analysis. Statistical Areas. 2023. Available online: <https://apps.bea.gov/regional/docs/msalist.cfm> (accessed on 3 September 2026).
50. Wood, S.N. Thin plate regression splines. *J. R. Stat. Soc. Ser. B Stat. Methodol.* **2003**, *65*, 95–114. [CrossRef]
51. Zou, H.; Hastie, T. Regularization and variable selection via the elastic net. *J. R. Stat. Soc. Ser. B Stat. Methodol.* **2005**, *67*, 301–320. [CrossRef]
52. Breiman, L. Random forests. *Mach. Learn.* **2001**, *45*, 5–32. [CrossRef]
53. Roberts, D.R.; Bahn, V.; Ciuti, S.; Boyce, M.S.; Elith, J.; Guillera-Arroita, G.; Hauenstein, S.; Lahoz-Monfort, J.J.; Schröder, B.; Thuiller, W.; et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* **2017**, *40*, 913–929. [CrossRef]
54. Ploton, P.; Mortier, F.; Réjou-Méchain, M.; Barbier, N.; Picard, N.; Rossi, V.; Dormann, C.; Cornu, G.; Viennois, G.; Bayol, N.; et al. Spatial validation reveals poor predictive performance of large-scale ecological mapping models. *Nat. Commun.* **2020**, *11*, 4540. [CrossRef]

55. Hand, D.J.; Till, R.J. A simple generalisation of the area under the ROC curve for multiple class classification problems. *Mach. Learn.* **2001**, *45*, 171–186. [\[CrossRef\]](#)
56. Fisher, A.; Rudin, C.; Dominici, F. All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously. *J. Mach. Learn. Res.* **2019**, *20*, 1–81.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.